2.25 Integration: Inverse Trigonometric Functions

Worked Solutions

1)

Let $x = \sin u$. Then

$$\frac{dx}{du} = \cos u, \qquad dx = \cos u \, du.$$

Therefore

$$\begin{aligned} I &= \int \frac{1}{\sqrt{1 - \sin^2 u}} \cos u \, du \\ &= \int \frac{\cos u}{\sqrt{\cos^2 u}} \, du = \int 1 \, du \\ &= u + c = \arcsin x + c. \end{aligned}$$

2)

Let $x = \cos u$. Then

$$\frac{dx}{du} = -\sin u, \qquad dx = -\sin u \, du.$$

Therefore

$$\begin{aligned} I &= \int \frac{1}{\sqrt{1 - \cos^2 u}} (-\sin u) \, du \\ &= -\int \frac{\sin u}{\sqrt{\sin^2 u}} \, du = -\int 1 \, du \\ &= -u + c = -\arccos x + c. \end{aligned}$$

3)

$$\begin{aligned} \int_{-1/2}^{1/2} \frac{1}{\sqrt{1 - x^2}} \, dx &= [\arcsin x]_{-1/2}^{1/2} \\ &= \arcsin \frac{1}{2} - \arcsin \left(-\frac{1}{2} \right) \\ &= \frac{\pi}{6} - \left(-\frac{\pi}{6} \right) = \frac{\pi}{3}. \end{aligned}$$

4)

Let $x = \tan u$. Then

$$\frac{dx}{du} = \sec^2 u, \qquad dx = \sec^2 u \, du.$$

Therefore

$$\begin{aligned} I &= \int \frac{\sec^2 u}{1 + \tan^2 u} \, du = \int 1 \, du \\ &= u + c = \arctan x + c. \end{aligned}$$

5)

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx = [\arctan x]_{-1}^{1}$$
$$= \arctan 1 - \arctan(-1) = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}.$$

6 a)

$$\int \frac{2}{\sqrt{1-x^2}}\,dx = 2\arcsin x + c.$$

6 b)

$$\int \frac{5}{1+x^2}\,dx = 5\arctan x + c.$$

6 c)

$$\int \frac{1}{\sqrt{1-9x^2}}\,dx = \int \frac{1}{\sqrt{1-(3x)^2}}\,dx$$
$$= \frac{1}{3}\arcsin 3x + c.$$

6 d)

$$\int \frac{1}{1+4x^2}\,dx = \int \frac{1}{1+(2x)^2}\,dx$$
$$= \frac{1}{2}\arctan 2x + c.$$

6 e)

$$\int \frac{1}{\sqrt{1-2x^2}}\,dx = \int \frac{1}{\sqrt{1-(\sqrt{2}x)^2}}\,dx$$
$$= \frac{1}{\sqrt{2}}\arcsin(\sqrt{2}x) + c$$
$$= \frac{\sqrt{2}}{2}\arcsin(\sqrt{2}x) + c.$$

6 f)

$$\int \frac{1}{1+8x^2}\,dx = \int \frac{1}{1+(2\sqrt{2}x)^2}\,dx$$
$$= \frac{1}{2\sqrt{2}}\arctan(2\sqrt{2}x) + c$$
$$= \frac{\sqrt{2}}{4}\arctan(2\sqrt{2}x) + c.$$

6 g)

$$\begin{aligned}\int \frac{10}{\sqrt{1-5x^2}}\,dx &= \int \frac{10}{\sqrt{1-(\sqrt{5}x)^2}}\,dx \\ &= \frac{10}{\sqrt{5}}\arcsin(\sqrt{5}x)+c \\ &= 2\sqrt{5}\arcsin(\sqrt{5}x)+c.\end{aligned}$$

6 h)

$$\begin{aligned}\int \frac{200}{1+10x^2}\,dx &= \int \frac{200}{1+(\sqrt{10}x)^2}\,dx \\ &= \frac{200}{\sqrt{10}}\arctan(\sqrt{10}x)+c \\ &= 20\sqrt{10}\arctan(\sqrt{10}x)+c.\end{aligned}$$

6 i)

$$\begin{aligned}\int \frac{4}{\sqrt{1-\frac{1}{4}x^2}}\,dx &= \int \frac{4}{\sqrt{1-(\frac{1}{2}x)^2}}\,dx \\ &= 4\times\frac{1}{1/2}\arcsin\left(\frac{1}{2}x\right)+c \\ &= 8\arcsin\left(\frac{1}{2}x\right)+c.\end{aligned}$$

6 j)

$$\begin{aligned}\int \frac{18}{1+\frac{2}{3}x^2}\,dx &= \int \frac{18}{1+(\sqrt{2/3}x)^2}\,dx \\ &= 18\sqrt{\frac{3}{2}}\arctan\left(\sqrt{\frac{2}{3}}x\right)+c \\ &= 9\sqrt{6}\arctan\left(\frac{\sqrt{6}}{3}x\right)+c.\end{aligned}$$

7 a)

$$\begin{aligned}\int_0^{\sqrt{2}/8} \frac{1}{\sqrt{1-16x^2}}\,dx &= \int_0^{\sqrt{2}/8} \frac{1}{\sqrt{1-(4x)^2}}\,dx \\ &= \left[\frac{1}{4}\arcsin(4x)\right]_0^{\sqrt{2}/8} \\ &= \frac{1}{4}\arcsin\left(4\times\frac{\sqrt{2}}{8}\right)-\frac{1}{4}\arcsin 0 \\ &= \frac{\pi}{16}-0=\frac{\pi}{16}.\end{aligned}$$

7 b)

$$\begin{aligned}\int_{1/(2\sqrt{3})}^{1/2} \frac{30}{\sqrt{1-3x^2}}\,dx &= \left[\frac{30}{\sqrt{3}}\arcsin(\sqrt{3}x)\right]_{1/(2\sqrt{3})}^{1/2} \\ &= [10\sqrt{3}\arcsin(\sqrt{3}x)]_{1/(2\sqrt{3})}^{1/2} \\ &= 10\sqrt{3}\arcsin\frac{\sqrt{3}}{2} - 10\sqrt{3}\arcsin\frac{1}{2} \\ &= 10\sqrt{3}\times\frac{\pi}{3} - 10\sqrt{3}\times\frac{\pi}{6} = \frac{5\sqrt{3}\pi}{3}.\end{aligned}$$

8 a)

$$\begin{aligned}\int_0^{\sqrt{3}/6} \frac{1}{1+36x^2}\,dx &= \left[\frac{1}{6}\arctan(6x)\right]_0^{\sqrt{3}/6} \\ &= \frac{1}{6}\arctan\sqrt{3} - 0 = \frac{\pi}{18}.\end{aligned}$$

8 b)

$$\begin{aligned}\int_{1/\sqrt{3}}^{1} \frac{12}{1+3x^2}\,dx &= \left[\frac{12}{\sqrt{3}}\arctan(\sqrt{3}x)\right]_{1/\sqrt{3}}^{1} \\ &= [4\sqrt{3}\arctan(\sqrt{3}x)]_{1/\sqrt{3}}^{1} \\ &= 4\sqrt{3}\arctan\sqrt{3} - 4\sqrt{3}\arctan 1 \\ &= 4\sqrt{3}\times\frac{\pi}{3} - 4\sqrt{3}\times\frac{\pi}{4} = \frac{\sqrt{3}\pi}{3}.\end{aligned}$$

9 a)

$$\frac{1}{2+x^2} \equiv \frac{1}{2(1+\frac{1}{2}x^2)} \equiv \frac{1}{2(1+(\frac{1}{\sqrt{2}}x)^2)}.$$

So $k = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

9 b)

$$\begin{aligned}\int \frac{1}{2+x^2}\,dx &= \frac{1}{2}\int \frac{1}{1+(\frac{1}{\sqrt{2}}x)^2}\,dx \\ &= \frac{1}{2}\times\frac{1}{1/\sqrt{2}}\arctan\left(\frac{1}{\sqrt{2}}x\right) + c \\ &= \frac{\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{2}x\right) + c.\end{aligned}$$

10 a)

$$\frac{1}{\sqrt{2-x^2}} \equiv \frac{1}{\sqrt{2}\sqrt{1-\frac{1}{2}x^2}} \equiv \frac{1}{\sqrt{2}\sqrt{1-(\frac{1}{\sqrt{2}}x)^2}}.$$

So $k = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

10 b)

$$\begin{aligned}\int \frac{1}{\sqrt{2-x^2}}\,dx &= \frac{1}{\sqrt{2}}\int \frac{1}{\sqrt{1-(\frac{1}{\sqrt{2}}x)^2}}\,dx \\ &= \frac{1}{\sqrt{2}} \times \frac{1}{1/\sqrt{2}} \arcsin\left(\frac{1}{\sqrt{2}}x\right) + c \\ &= \arcsin\left(\frac{\sqrt{2}}{2}x\right) + c.\end{aligned}$$

11 a)

$$\begin{aligned}\int \frac{1}{4+x^2}\,dx &= \frac{1}{4}\int \frac{1}{1+(\frac{1}{2}x)^2}\,dx \\ &= \frac{1}{2}\arctan\left(\frac{1}{2}x\right) + c.\end{aligned}$$

11 b)

$$\begin{aligned}\int \frac{1}{\sqrt{9-x^2}}\,dx &= \frac{1}{3}\int \frac{1}{\sqrt{1-(\frac{1}{3}x)^2}}\,dx \\ &= \arcsin\left(\frac{1}{3}x\right) + c.\end{aligned}$$

11 c)

$$\begin{aligned}\int \frac{1}{5+x^2}\,dx &= \frac{1}{5}\int \frac{1}{1+(\frac{1}{\sqrt{5}}x)^2}\,dx \\ &= \frac{\sqrt{5}}{5}\arctan\left(\frac{\sqrt{5}}{5}x\right) + c.\end{aligned}$$

11 d)

$$\begin{aligned}\int \frac{1}{\sqrt{8-x^2}}\,dx &= \frac{1}{2\sqrt{2}}\int \frac{1}{\sqrt{1-(\frac{1}{2\sqrt{2}}x)^2}}\,dx \\ &= \arcsin\left(\frac{\sqrt{2}}{4}x\right) + c.\end{aligned}$$

11 e)

$$\begin{aligned}\int \frac{4}{12+x^2}\,dx &= \frac{1}{3}\int \frac{1}{1+(\frac{1}{2\sqrt{3}}x)^2}\,dx \\ &= \frac{2\sqrt{3}}{3}\arctan\left(\frac{\sqrt{3}}{6}x\right)+c.\end{aligned}$$

11 f)

$$\begin{aligned}\int \frac{10}{\sqrt{5-x^2}}\,dx &= 2\sqrt{5}\int \frac{1}{\sqrt{1-(\frac{1}{\sqrt{5}}x)^2}}\,dx \\ &= 10\arcsin\left(\frac{\sqrt{5}}{5}x\right)+c.\end{aligned}$$

11 g)

$$\begin{aligned}\int \frac{1}{2+4x^2}\,dx &= \frac{1}{2}\int \frac{1}{1+(\sqrt{2}x)^2}\,dx \\ &= \frac{\sqrt{2}}{4}\arctan(\sqrt{2}x)+c.\end{aligned}$$

11 h)

$$\begin{aligned}\int \frac{1}{\sqrt{16-9x^2}}\,dx &= \frac{1}{4}\int \frac{1}{\sqrt{1-(\frac{3}{4}x)^2}}\,dx \\ &= \frac{1}{3}\arcsin\left(\frac{3}{4}x\right)+c.\end{aligned}$$

11 i)

$$\begin{aligned}\int \frac{2}{4+25x^2}\,dx &= \frac{1}{2}\int \frac{1}{1+(\frac{5}{2}x)^2}\,dx \\ &= \frac{1}{5}\arctan\left(\frac{5}{2}x\right)+c.\end{aligned}$$

11 j)

$$\begin{aligned}\int \frac{3}{\sqrt{12-\frac{1}{3}x^2}}\,dx &= \frac{\sqrt{3}}{2}\int \frac{1}{\sqrt{1-(\frac{1}{6}x)^2}}\,dx \\ &= 3\sqrt{3}\arcsin\left(\frac{1}{6}x\right)+c.\end{aligned}$$

12 a)

$$\begin{aligned}\int_{-\sqrt{3}/3}^{\sqrt{3}/3} \frac{1}{\sqrt{4-9x^2}}\,dx &= \frac{1}{2}\int_{-\sqrt{3}/3}^{\sqrt{3}/3} \frac{1}{\sqrt{1-(\frac{3}{2}x)^2}}\,dx \\ &= \left[\frac{1}{3}\arcsin\left(\frac{3}{2}x\right)\right]_{-\sqrt{3}/3}^{\sqrt{3}/3} \\ &= \frac{1}{3}\arcsin\frac{\sqrt{3}}{2} - \frac{1}{3}\arcsin\left(-\frac{\sqrt{3}}{2}\right) \\ &= \frac{\pi}{9} - \left(-\frac{\pi}{9}\right) = \frac{2\pi}{9}.\end{aligned}$$

12 b)

$$\begin{aligned}\int_{0}^{\frac{5}{8}(\sqrt{5}-1)} \frac{3}{\sqrt{100-16x^2}}\,dx &= \frac{3}{10}\int_{0}^{\frac{5}{8}(\sqrt{5}-1)} \frac{1}{\sqrt{1-(\frac{2}{5}x)^2}}\,dx \\ &= \left[\frac{3}{4}\arcsin\left(\frac{2}{5}x\right)\right]_{0}^{\frac{5}{8}(\sqrt{5}-1)} \\ &= \frac{3}{4}\arcsin\left(\frac{1}{4}(\sqrt{5}-1)\right) - 0 \\ &= \frac{3\pi}{40}.\end{aligned}$$

13 a)

$$\begin{aligned}\int_{-5\sqrt{3}/27}^{5\sqrt{3}/27} \frac{1}{25+81x^2}\,dx &= \frac{1}{25}\int_{-5\sqrt{3}/27}^{5\sqrt{3}/27} \frac{1}{1+(\frac{9}{5}x)^2}\,dx \\ &= \left[\frac{1}{45}\arctan\left(\frac{9}{5}x\right)\right]_{-5\sqrt{3}/27}^{5\sqrt{3}/27} \\ &= \frac{1}{45}\arctan\frac{\sqrt{3}}{3} - \frac{1}{45}\arctan\left(-\frac{\sqrt{3}}{3}\right) \\ &= \frac{\pi}{270} - \left(-\frac{\pi}{270}\right) = \frac{\pi}{135}.\end{aligned}$$

13 b)

$$\begin{aligned}\int_{0}^{3(\sqrt{2}-\sqrt{3/2})} \frac{9}{9+2x^2}\,dx &= \int_{0}^{3(\sqrt{2}-\sqrt{3/2})} \frac{1}{1+(\frac{\sqrt{2}}{3}x)^2}\,dx \\ &= \left[\frac{3\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{3}x\right)\right]_{0}^{3(\sqrt{2}-\sqrt{3/2})} \\ &= \frac{\sqrt{2}}{8}\pi = \frac{\pi\sqrt{2}}{8}.\end{aligned}$$

14 a)

$$\begin{aligned}\int \frac{1}{\sqrt{a^2-x^2}}\,dx &= \frac{1}{a}\int \frac{1}{\sqrt{1-(\frac{x}{a})^2}}\,dx \\ &= \arcsin\left(\frac{x}{a}\right)+c.\end{aligned}$$

14 b)

$$\begin{aligned}\int \frac{1}{a^2+x^2}\,dx &= \frac{1}{a^2}\int \frac{1}{1+(\frac{x}{a})^2}\,dx \\ &= \frac{1}{a}\arctan\left(\frac{x}{a}\right)+c.\end{aligned}$$

14 c)

$$\begin{aligned}\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx &= \frac{1}{a}\int \frac{1}{\sqrt{1-(\frac{b}{a}x)^2}}\,dx \\ &= \frac{1}{b}\arcsin\left(\frac{b}{a}x\right)+c.\end{aligned}$$

14 d)

$$\begin{aligned}\int \frac{1}{a^2+b^2x^2}\,dx &= \frac{1}{a^2}\int \frac{1}{1+(\frac{b}{a}x)^2}\,dx \\ &= \frac{1}{ab}\arctan\left(\frac{b}{a}x\right)+c.\end{aligned}$$

15)

$$\begin{aligned}\int_0^{\sqrt{ab}} \frac{1}{\sqrt{b-\frac{1}{a}x^2}}\,dx &= \frac{1}{\sqrt{b}}\int_0^{\sqrt{ab}} \frac{1}{\sqrt{1-(\frac{x}{\sqrt{ab}})^2}}\,dx \\ &= \left[\sqrt{a}\arcsin\left(\frac{x}{\sqrt{ab}}\right)\right]_0^{\sqrt{ab}} \\ &= \sqrt{a}\arcsin 1-\sqrt{a}\arcsin 0 = \frac{\pi\sqrt{a}}{2}.\end{aligned}$$

16)

$$\int_{\sqrt{6}/2}^{\sqrt{3}} \frac{6}{\sqrt{18-3x^2}}\,dx = \sqrt{2}\int_{\sqrt{6}/2}^{\sqrt{3}} \frac{1}{\sqrt{1-(\frac{1}{\sqrt{6}}x)^2}}\,dx$$

$$= \left[2\sqrt{3}\arcsin\left(\frac{\sqrt{6}}{6}x\right)\right]_{\sqrt{3}}^{\sqrt{6}/2}$$
$$= 2\sqrt{3}\arcsin\frac{\sqrt{2}}{2} - 2\sqrt{3}\arcsin\frac{1}{2}$$
$$= \frac{\sqrt{3}\pi}{2} - \frac{\sqrt{3}\pi}{3} = \frac{\pi\sqrt{3}}{6}\ \text{units}^2.$$

17)

$$\int_{3\sqrt{3}-6}^{6-3\sqrt{3}} \frac{9}{36+4x^2}\,dx = \frac{1}{4}\int_{3\sqrt{3}-6}^{6-3\sqrt{3}} \frac{1}{1+(\frac{1}{3}x)^2}\,dx$$
$$= \left[\frac{3}{4}\arctan\left(\frac{1}{3}x\right)\right]_{3\sqrt{3}-6}^{6-3\sqrt{3}}$$
$$= \frac{\pi}{16} - \left(-\frac{\pi}{16}\right) = \frac{\pi}{8}\ \text{units}^2.$$

18 a)

$$\frac{19x^2+34}{(1+x^2)(9+4x^2)} \equiv \frac{Ax+B}{1+x^2} + \frac{Cx+D}{9+4x^2},$$
$$19x^2+34 \equiv (Ax+B)(9+4x^2) + (Cx+D)(1+x^2).$$

Let $x = 0$: $34 = 9B + D$. Let $x = 1$: $53 = 13A + 13B + 2C + 2D$. Let $x = -1$: $53 = -13A + 13B - 2C + 2D$. Let $x = 2$: $110 = 50A + 25B + 10C + 5D$. Solving gives $A = 0$, $B = 3$, $C = 0$ and $D = 7$. Therefore

$$\frac{19x^2+34}{(1+x^2)(9+4x^2)} \equiv \frac{3}{1+x^2} + \frac{7}{9+4x^2}.$$

18 b)

$$\int \frac{19x^2+34}{(1+x^2)(9+4x^2)}\,dx = \int \left(\frac{3}{1+x^2} + \frac{7}{9(1+(\frac{2}{3}x)^2)}\right)dx$$
$$= 3\arctan x + \frac{7}{9}\times\frac{3}{2}\arctan\left(\frac{2}{3}x\right) + c$$
$$= 3\arctan x + \frac{7}{6}\arctan\left(\frac{2}{3}x\right) + c.$$

19 a)

$$\frac{107x^2-3}{(4+9x^2)(1+25x^2)} \equiv \frac{Ax+B}{4+9x^2} + \frac{Cx+D}{1+25x^2},$$
$$107x^2-3 \equiv (Ax+B)(1+25x^2) + (Cx+D)(4+9x^2).$$

Let $x = 0$: $-3 = B + 4D$. Let $x = 1$: $104 = 26A + 26B + 13C + 13D$. Let $x = -1$: $104 = -26A + 26B - 13C + 13D$. Let $x = 2$: $425 = 202A + 101B + 80C + 40D$. Solving gives $A = 0$, $B = 5$, $C = 0$ and $D = -2$, so

$$\frac{107x^2 - 3}{(4 + 9x^2)(1 + 25x^2)} \equiv \frac{5}{4 + 9x^2} - \frac{2}{1 + 25x^2}.$$

Hence

$$\int \frac{107x^2 - 3}{(4 + 9x^2)(1 + 25x^2)}\, dx = \frac{5}{6} \arctan\left(\frac{3}{2}x\right) - \frac{2}{5} \arctan(5x) + c.$$

19 b)

$$\frac{8x^4 + 12x^3 + 33x^2 + 3x + 4}{(4 + x^2)(1 + 4x^2)} \equiv E + \frac{Ax + B}{4 + x^2} + \frac{Cx + D}{1 + 4x^2}.$$

The coefficient of x^4 in the numerator is 8 and in the denominator is 4, so $E = 2$. Thus

$$\begin{aligned} 8x^4 + 12x^3 + 33x^2 + 3x + 4 \equiv 2(4 + x^2)(1 + 4x^2) + (Ax + B)(1 + 4x^2) \\ + (Cx + D)(4 + x^2). \end{aligned}$$

Let $x = 0$: $B + D = -4$. Let $x = 1$: $A + B + C + D = 2$. Let $x = -1$: $A - B + C - D = 4$. Let $x = 2$: $34A + 17B + 16C + 8D = 94$. Solving gives $A = 3$, $B = 0$, $C = 0$, $D = -1$. Therefore

$$\frac{8x^4 + 12x^3 + 33x^2 + 3x + 4}{(4 + x^2)(1 + 4x^2)} \equiv 2 + \frac{3x}{4 + x^2} - \frac{1}{1 + 4x^2}.$$

Hence

$$\int \frac{8x^4 + 12x^3 + 33x^2 + 3x + 4}{(4 + x^2)(1 + 4x^2)}\, dx = 2x + \frac{3}{2} \ln|4 + x^2| - \frac{1}{2} \arctan 2x + c.$$

20)

$$\begin{aligned} \frac{16x^2 + 5x + 4}{x(8x^2 + 2)} &\equiv \frac{A}{x} + \frac{Bx + C}{8x^2 + 2}, \\ 16x^2 + 5x + 4 &\equiv A(8x^2 + 2) + (Bx + C)x. \end{aligned}$$

Let $x = 0$: $4 = 2A$, so $A = 2$. Let $x = 1$: $B + C = 5$. Let $x = -1$: $B - C = -5$. Thus $B = 0$ and $C = 5$, so

$$\frac{16x^2 + 5x + 4}{x(8x^2 + 2)} \equiv \frac{2}{x} + \frac{5}{8x^2 + 2}.$$

Therefore

$$\begin{aligned} \int_{1/2}^{\sqrt{3}/2} \frac{16x^2 + 5x + 4}{x(8x^2 + 2)}\, dx &= \left[2 \ln|x| + \frac{5}{4} \arctan(2x)\right]_{1/2}^{\sqrt{3}/2} \\ &= \left(\ln \frac{3}{4} + \frac{5\pi}{12}\right) - \left(\ln \frac{1}{4} + \frac{5\pi}{16}\right) \\ &= \ln 3 + \frac{5\pi}{48}. \end{aligned}$$

So $a = \frac{5}{48}$ and $b = 3$.

21)

$$\begin{aligned}\int \frac{1+x}{1+x^2}\,dx &= \int \left(\frac{1}{1+x^2}+\frac{x}{1+x^2}\right)dx \\ &= \arctan x + \frac{1}{2}\ln|1+x^2| + c.\end{aligned}$$

22 a)

$$\begin{aligned}\int \frac{2x+1}{1+4x^2}\,dx &= \int \left(\frac{2x}{1+4x^2}+\frac{1}{1+(2x)^2}\right)dx \\ &= \frac{1}{4}\ln|1+4x^2| + \frac{1}{2}\arctan 2x + c.\end{aligned}$$

22 b)

$$\begin{aligned}\int \frac{3+5x}{4+x^2}\,dx &= \int \left(\frac{3}{4(1+(\frac{1}{2}x)^2)}+\frac{5x}{4+x^2}\right)dx \\ &= \frac{3}{2}\arctan\left(\frac{1}{2}x\right) + \frac{5}{2}\ln|4+x^2| + c.\end{aligned}$$

22 c)

$$\begin{aligned}\int \frac{2-3x}{9+4x^2}\,dx &= \int \left(\frac{2}{9(1+(\frac{2}{3}x)^2)}-\frac{3x}{9+4x^2}\right)dx \\ &= \frac{1}{3}\arctan\left(\frac{2}{3}x\right) - \frac{3}{8}\ln|9+4x^2| + c.\end{aligned}$$

22 d)

$$\begin{aligned}\int \frac{5-4x}{25+16x^2}\,dx &= \int \left(\frac{1}{5(1+(\frac{4}{5}x)^2)}-\frac{4x}{25+16x^2}\right)dx \\ &= \frac{1}{4}\arctan\left(\frac{4}{5}x\right) - \frac{1}{8}\ln|25+16x^2| + c.\end{aligned}$$

23)

$$\begin{aligned}\int \frac{1+x}{\sqrt{1-x^2}}\,dx &= \int \left(\frac{1}{\sqrt{1-x^2}}+\frac{x}{\sqrt{1-x^2}}\right)dx \\ &= \arcsin x - \sqrt{1-x^2} + c.\end{aligned}$$

24 a)

$$\begin{aligned}\int \frac{3x-1}{\sqrt{1-9x^2}}\,dx &= \int \left(\frac{3x}{\sqrt{1-9x^2}} - \frac{1}{\sqrt{1-(3x)^2}}\right) dx \\ &= -\frac{1}{3}\sqrt{1-9x^2} - \frac{1}{3}\arcsin 3x + c.\end{aligned}$$

24 b)

$$\begin{aligned}\int \frac{1+8x}{\sqrt{4-x^2}}\,dx &= \int \left(\frac{1}{2\sqrt{1-(\frac{1}{2}x)^2}} + \frac{8x}{\sqrt{4-x^2}}\right) dx \\ &= \arcsin\left(\frac{1}{2}x\right) - 8\sqrt{4-x^2} + c.\end{aligned}$$

24 c)

$$\begin{aligned}\int \frac{6-x}{\sqrt{9-4x^2}}\,dx &= \int \left(\frac{2}{\sqrt{1-(\frac{2}{3}x)^2}} - \frac{x}{\sqrt{9-4x^2}}\right) dx \\ &= 3\arcsin\left(\frac{2}{3}x\right) + \frac{1}{4}\sqrt{9-4x^2} + c.\end{aligned}$$

24 d)

$$\begin{aligned}\int \frac{10-9x}{\sqrt{4-25x^2}}\,dx &= \int \left(\frac{5}{\sqrt{1-(\frac{5}{2}x)^2}} - \frac{9x}{\sqrt{4-25x^2}}\right) dx \\ &= 2\arcsin\left(\frac{5}{2}x\right) + \frac{9}{25}\sqrt{4-25x^2} + c.\end{aligned}$$

25)

$$\begin{aligned}\int_{-3}^{3} \frac{3+x}{9+x^2}\,dx &= \int_{-3}^{3} \left(\frac{3}{9+x^2} + \frac{x}{9+x^2}\right) dx \\ &= \left[\arctan\left(\frac{1}{3}x\right) + \frac{1}{2}\ln|9+x^2|\right]_{-3}^{3} \\ &= \left(\frac{\pi}{4} + \frac{1}{2}\ln 18\right) - \left(-\frac{\pi}{4} + \frac{1}{2}\ln 18\right) \\ &= \frac{\pi}{2}\ \text{units}^2.\end{aligned}$$

26 a)

$$\begin{aligned}\frac{dy}{dx} &= \frac{x+1}{\sqrt{9-x^2}} = \frac{x}{\sqrt{9-x^2}} + \frac{1}{\sqrt{9-x^2}} \\ &= \frac{x}{\sqrt{9-x^2}} + \frac{1}{3\sqrt{1-(\frac{1}{3}x)^2}}.\end{aligned}$$

Integrating,

$$y = -\sqrt{9-x^2} + \arcsin\left(\frac{1}{3}x\right) + c.$$

26 b)

Let $x = 0$, $y = 3$:

$$3 = -3 + \arcsin 0 + c \quad \Rightarrow \quad c = 6.$$

Therefore $y = -\sqrt{9-x^2} + \arcsin\left(\frac{1}{3}x\right) + 6$.

27 a)

Use integration by substitution. Let $u = 25 - x^2$, so $\frac{du}{dx} = -2x$ and $dx = -\frac{1}{2x}\,du$. Then

$$\begin{aligned}\int \frac{x}{\sqrt{25-x^2}}\,dx &= \int \frac{x}{\sqrt{u}}\left(-\frac{1}{2x}\right)du \\ &= -\frac{1}{2}\int u^{-1/2}\,du = -u^{1/2} + c \\ &= -\sqrt{25-x^2} + c.\end{aligned}$$

27 b)

$$\begin{aligned}\int \frac{1}{\sqrt{25-x^2}}\,dx &= \frac{1}{5}\int \frac{1}{\sqrt{1-(\frac{1}{5}x)^2}}\,dx \\ &= \arcsin\left(\frac{1}{5}x\right) + c.\end{aligned}$$

27 c)

The mean value is

$$\begin{aligned}\frac{1}{0-(-5/2)}\int_{-5/2}^{0} \frac{5x+20}{\sqrt{25-x^2}}\,dx &= 2\int_{-5/2}^{0} \frac{x+4}{\sqrt{25-x^2}}\,dx \\ &= 2\left[-\sqrt{25-x^2} + 4\arcsin\left(\frac{1}{5}x\right)\right]_{-5/2}^{0} \\ &= 2\left((-5) - \left(-\frac{5\sqrt{3}}{2} - \frac{2\pi}{3}\right)\right) \\ &= -10 + 5\sqrt{3} + \frac{4\pi}{3}.\end{aligned}$$

28)

$$\begin{aligned}\int_{-\sqrt{2}/3}^{1/3} \frac{4-9x}{\sqrt{4-9x^2}}\,dx &= \int_{-\sqrt{2}/3}^{1/3} \left(\frac{4}{\sqrt{4-9x^2}} - \frac{9x}{\sqrt{4-9x^2}}\right) dx \\ &= \left[\frac{4}{3}\arcsin\left(\frac{3}{2}x\right) + \sqrt{4-9x^2}\right]_{-\sqrt{2}/3}^{1/3} \\ &= \left(\frac{2\pi}{9} + \sqrt{3}\right) - \left(-\frac{\pi}{3} + \sqrt{2}\right) \\ &= \frac{5\pi}{9} + \sqrt{3} - \sqrt{2} \text{ units}^2.\end{aligned}$$

29 a)

Let $x = \sin u$, so $dx = \cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{1-x^2}}\,dx &= \int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \int 1\,du \\ &= u + c = \arcsin x + c.\end{aligned}$$

So $x = \sin u$ gives $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c$.

29 b)

Let $x = \tan u$, so $dx = \sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{1+x^2}\,dx &= \int \frac{\sec^2 u}{1+\tan^2 u}\,du = \int 1\,du \\ &= u + c = \arctan x + c.\end{aligned}$$

So $x = \tan u$ gives $\displaystyle\int \frac{1}{1+x^2}\,dx = \arctan x + c$.

29 c)

Let $x = \frac{1}{2}\sin u$, so $dx = \frac{1}{2}\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{1-4x^2}}\,dx &= \frac{1}{2}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \frac{1}{2}u + c \\ &= \frac{1}{2}\arcsin(2x) + c.\end{aligned}$$

29 d)

Let $x = \frac{1}{3}\tan u$, so $dx = \frac{1}{3}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{1+9x^2}\,dx &= \frac{1}{3}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{3}u + c \\ &= \frac{1}{3}\arctan(3x) + c.\end{aligned}$$

29 e)

Let $x = 3\sin u$, so $dx = 3\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{9-x^2}}\,dx &= \int \frac{3\cos u}{\sqrt{9-9\sin^2 u}}\,du = \int 1\,du \\ &= u + c = \arcsin\left(\frac{x}{3}\right) + c.\end{aligned}$$

29 f)

Let $x = 2\tan u$, so $dx = 2\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{4+x^2}\,dx &= \frac{1}{2}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{2}u + c \\ &= \frac{1}{2}\arctan\left(\frac{x}{2}\right) + c.\end{aligned}$$

29 g)

Let $x = \frac{1}{5}\sin u$, so $dx = \frac{1}{5}\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{1-25x^2}}\,dx &= \frac{1}{5}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \frac{1}{5}u + c \\ &= \frac{1}{5}\arcsin(5x) + c.\end{aligned}$$

29 h)

Let $x = \frac{1}{9}\tan u$, so $dx = \frac{1}{9}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{1+81x^2}\,dx &= \frac{1}{9}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{9}u + c \\ &= \frac{1}{9}\arctan(9x) + c.\end{aligned}$$

29 i)

Let $x = \frac{2}{3}\sin u$, so $dx = \frac{2}{3}\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{4-9x^2}}\,dx &= \frac{1}{3}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \frac{1}{3}u + c \\ &= \frac{1}{3}\arcsin\left(\frac{3}{2}x\right) + c.\end{aligned}$$

29 j)

Let $x = \frac{3}{2}\tan u$, so $dx = \frac{3}{2}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{9+4x^2}\,dx &= \frac{1}{6}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{6}u + c \\ &= \frac{1}{6}\arctan\left(\frac{2}{3}x\right) + c.\end{aligned}$$

29 k)

Let $x = \frac{5}{2}\sin u$, so $dx = \frac{5}{2}\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{25-4x^2}}\,dx &= \frac{1}{2}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \frac{1}{2}u + c \\ &= \frac{1}{2}\arcsin\left(\frac{2}{5}x\right) + c.\end{aligned}$$

29 l)

Let $x = \frac{7}{6}\tan u$, so $dx = \frac{7}{6}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{49+36x^2}\,dx &= \frac{1}{42}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{42}u + c \\ &= \frac{1}{42}\arctan\left(\frac{6}{7}x\right) + c.\end{aligned}$$

29 m)

Let $x = \frac{4}{\sqrt{3}}\sin u$, so $dx = \frac{4}{\sqrt{3}}\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{16-3x^2}}\,dx &= \frac{1}{\sqrt{3}}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \frac{1}{\sqrt{3}}u + c \\ &= \frac{1}{\sqrt{3}}\arcsin\left(\frac{\sqrt{3}}{4}x\right) + c.\end{aligned}$$

29 n)

Let $x = \frac{2}{\sqrt{5}}\tan u$, so $dx = \frac{2}{\sqrt{5}}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{4+5x^2}\,dx &= \frac{1}{2\sqrt{5}}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{2\sqrt{5}}u + c \\ &= \frac{1}{2\sqrt{5}}\arctan\left(\frac{\sqrt{5}}{2}x\right) + c.\end{aligned}$$

29 o)

Let $x = \frac{2\sqrt{2}}{9}\sin u$, so $dx = \frac{2\sqrt{2}}{9}\cos u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{8-81x^2}}\,dx &= \frac{1}{9}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du = \frac{1}{9}u + c \\ &= \frac{1}{9}\arcsin\left(\frac{9x}{2\sqrt{2}}\right) + c.\end{aligned}$$

29 p)

Let $x = \frac{\sqrt{12}}{5}\tan u$, so $dx = \frac{\sqrt{12}}{5}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{12+25x^2}\,dx &= \frac{1}{5\sqrt{12}}\int \frac{\sec^2 u}{1+\tan^2 u}\,du = \frac{1}{5\sqrt{12}}u + c \\ &= \frac{1}{10\sqrt{3}}\arctan\left(\frac{5x}{\sqrt{12}}\right) + c.\end{aligned}$$

29 q)

Let $x = \sqrt{\frac{a}{b}} \sin u$, so $dx = \sqrt{\frac{a}{b}} \cos u \, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{a - bx^2}} \, dx &= \frac{1}{\sqrt{b}} \int \frac{\sqrt{a/b} \cos u}{\sqrt{1 - \sin^2 u}} \, du \\ &= \frac{1}{\sqrt{b}} u + c = \frac{1}{\sqrt{b}} \arcsin\left(\sqrt{\frac{b}{a}} x\right) + c.\end{aligned}$$

29 r)

Let $x = \sqrt{\frac{a}{b}} \tan u$, so $dx = \sqrt{\frac{a}{b}} \sec^2 u \, du$. Then

$$\begin{aligned}\int \frac{1}{a + bx^2} \, dx &= \frac{1}{\sqrt{ab}} \int \frac{\sec^2 u}{1 + \tan^2 u} \, du \\ &= \frac{1}{\sqrt{ab}} u + c = \frac{1}{\sqrt{ab}} \arctan\left(\sqrt{\frac{b}{a}} x\right) + c.\end{aligned}$$

30 a)

Let $x = \sin u$, so $dx = \cos u \, du$. Then

$$\begin{aligned}\int \sqrt{1 - x^2} \, dx &= \int \cos^2 u \, du = \frac{1}{2} u + \frac{1}{2} \sin u \cos u + c \\ &= \frac{1}{2} \arcsin x + \frac{1}{2} x \sqrt{1 - x^2} + c.\end{aligned}$$

30 b)

Let $x = \sin u$, so $dx = \cos u \, du$. Then

$$\begin{aligned}\int (1 - x^2)^{3/2} \, dx &= \int \cos^4 u \, du \\ &= \frac{3}{8} u + \frac{1}{4} \sin 2u + \frac{1}{32} \sin 4u + c \\ &= \frac{1}{8} \left(3 \arcsin x + x \sqrt{1 - x^2}(5 - 2x^2)\right) + c.\end{aligned}$$

31 a)

Let $x = \tan u$, so $dx = \sec^2 u \, du$. Then

$$\begin{aligned}\int \frac{1}{(1 + x^2)^2} \, dx &= \int \frac{\sec^2 u}{\sec^4 u} \, du = \int \cos^2 u \, du \\ &= \frac{1}{2} u + \frac{1}{2} \sin u \cos u + c \\ &= \frac{1}{2} \arctan x + \frac{x}{2(1 + x^2)} + c.\end{aligned}$$

31 b)

Let $x = 2\tan u$, so $dx = 2\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{(4+x^2)^2}\,dx &= \frac{1}{8}\int \cos^2 u\,du \\ &= \frac{1}{16}u + \frac{1}{16}\sin u\cos u + c \\ &= \frac{1}{16}\arctan\left(\frac{x}{2}\right) + \frac{x}{8(4+x^2)} + c.\end{aligned}$$

31 c)

Let $x = \tan u$, so $dx = \sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{(1+x^2)^3}\,dx &= \int \cos^4 u\,du \\ &= \frac{3}{8}u + \frac{1}{4}\sin 2u + \frac{1}{32}\sin 4u + c \\ &= \frac{3}{8}\arctan x + \frac{x(5+3x^2)}{8(1+x^2)^2} + c.\end{aligned}$$

31 d)

Let $2x = 3\tan u$, so $dx = \frac{3}{2}\sec^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{(9+4x^2)^3}\,dx &= \frac{1}{486}\int \cos^4 u\,du \\ &= \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + \frac{x(15+4x^2)}{216(9+4x^2)^2} + c.\end{aligned}$$

32 a)

Let $x = 6\sin u$, so $dx = 6\cos u\,du$. Then

$$\begin{aligned}\int \sqrt{36-x^2}\,dx &= 36\int \cos^2 u\,du \\ &= 18u + 18\sin u\cos u + c \\ &= 18\arcsin\left(\frac{x}{6}\right) + \frac{x}{2}\sqrt{36-x^2} + c.\end{aligned}$$

32 b)

Let $2x = \sin u$, so $dx = \frac{1}{2}\cos u\,du$. Then

$$\begin{aligned}\int (1-4x^2)^{3/2}\,dx &= \frac{1}{2}\int \cos^4 u\,du \\ &= \frac{3}{16}u + \frac{1}{8}\sin 2u + \frac{1}{64}\sin 4u + c \\ &= \frac{1}{16}\left(3\arcsin(2x) + 2x\sqrt{1-4x^2}(5-8x^2)\right) + c.\end{aligned}$$

33 a)

$$26 - 10x + x^2 = (x-5)^2 + 1.$$

33 b)

$$101 + 20x + x^2 = (x+10)^2 + 1.$$

33 c)

$$2 + 4x + 4x^2 = (2x+1)^2 + 1.$$

33 d)

$$5 - 12x + 9x^2 = (3x-2)^2 + 1.$$

33 e)

$$25 - 8x + x^2 = (x-4)^2 + 9.$$

33 f)

$$29 + 8x + 4x^2 = (2x+1)^2 + 28.$$

34 a)

$$2 + 2x + x^2 = (x+1)^2 + 1,$$

so $p = 1$ and $q = 1$.

34 b)

Using $x + 1 = \tan u$, $dx = \sec^2 u \, du$:

$$\begin{aligned}\int \frac{1}{2 + 2x + x^2}\, dx &= \int \frac{\sec^2 u}{1 + \tan^2 u}\, du = u + c \\ &= \arctan(x+1) + c.\end{aligned}$$

35 a)

$$\begin{aligned}\int \frac{1}{10 - 6x + x^2}\, dx &= \int \frac{1}{1 + (x-3)^2}\, dx \\ &= \arctan(x-3) + c.\end{aligned}$$

35 b)

$$\int \frac{1}{65+16x+x^2}\,dx = \int \frac{1}{1+(x+8)^2}\,dx$$
$$= \arctan(x+8)+c.$$

36 a)

$$10+18x+9x^2 = (3x+3)^2+1,$$

so $p=1$, $q=1$ and $r=3$.

36 b)

Using $3x+3=\tan u$, $3\,dx=\sec^2 u\,du$:

$$\int \frac{1}{10+18x+9x^2}\,dx = \frac{1}{3}\int \frac{\sec^2 u}{1+\tan^2 u}\,du$$
$$= \frac{1}{3}u+c = \frac{1}{3}\arctan(3x+3)+c.$$

37 a)

$$\int \frac{1}{10+12x+4x^2}\,dx = \frac{1}{4}\int \frac{1}{1+(x+\frac{3}{2})^2}\,dx$$
$$= \frac{1}{2}\arctan(2x+3)+c.$$

37 b)

$$\int \frac{1}{50-28x+4x^2}\,dx = \frac{1}{4}\int \frac{1}{1+(x-\frac{7}{2})^2}\,dx$$
$$= \frac{1}{2}\arctan(2x-7)+c.$$

38 a)

$$13-6x+x^2 = (x-3)^2+4,$$

so $p=4$ and $q=-3$.

38 b)

Using $\frac{x-3}{2}=\tan u$, $dx=2\sec^2 u\,du$:

$$\int \frac{1}{13-6x+x^2}\,dx = \frac{1}{2}\int \frac{\sec^2 u}{1+\tan^2 u}\,du$$
$$= \frac{1}{2}u+c = \frac{1}{2}\arctan\left(\frac{x-3}{2}\right)+c.$$

39 a)

$$\begin{aligned}\int \frac{1}{26+8x+2x^2}\,dx &= \frac{1}{2}\int \frac{1}{(x+2)^2+9}\,dx \\ &= \frac{1}{6}\arctan\left(\frac{x+2}{3}\right)+c.\end{aligned}$$

39 b)

$$\begin{aligned}\int \frac{6}{5+12x+8x^2}\,dx &= \frac{3}{4}\int \frac{1}{(x+\frac{3}{4})^2+(\frac{1}{4})^2}\,dx \\ &= 3\arctan(4x+3)+c.\end{aligned}$$

40)

$$\begin{aligned}\int_{7/3}^{3} \frac{2}{65-48x+9x^2}\,dx &= \frac{2}{3}\int_{7/3}^{3} \frac{1}{1+(3x-8)^2}\,dx \\ &= \left[\frac{2}{3}\arctan(3x-8)\right]_{7/3}^{3} \\ &= \frac{2}{3}\left(\frac{\pi}{4}-\left(-\frac{\pi}{4}\right)\right) = \frac{\pi}{3}\ \text{units}^2.\end{aligned}$$

41)

$$\begin{aligned}\int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{7}{41+10x+x^2}\,dx &= 7\int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{1}{16+(x+5)^2}\,dx \\ &= \left[\frac{7}{4}\arctan\left(\frac{x+5}{4}\right)\right]_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \\ &= \frac{7}{4}\left(\frac{\pi}{3}-\left(-\frac{\pi}{3}\right)\right) = \frac{7\pi}{6}\ \text{units}^2.\end{aligned}$$

42 a)

$$-8+6x-x^2 = 1-(x-3)^2.$$

42 b)

$$-48-14x-x^2 = 1-(x+7)^2.$$

42 c)

$$-15+16x-4x^2 = 1-4(x-2)^2.$$

42 d)

$$-224 - 90x - 9x^2 = 1 - \left(3x + \frac{15}{2}\right)^2.$$

42 e)

$$-32 + 12x - x^2 = 4 - (x - 6)^2.$$

42 f)

$$-128 - 48x - 4x^2 = 16 - (2x + 6)^2.$$

43 a)

$$-63 + 16x - x^2 = 1 - (x - 8)^2,$$

so $p = 1$ and $q = -8$.

43 b)

Using $x - 8 = \sin u$, $dx = \cos u\, du$:

$$\begin{aligned}\int \frac{1}{\sqrt{-63 + 16x - x^2}}\, dx &= \int \frac{\cos u}{\sqrt{1 - \sin^2 u}}\, du \\ &= u + c = \arcsin(x - 8) + c.\end{aligned}$$

44 a)

$$\begin{aligned}\int \frac{1}{\sqrt{-3 - 4x - x^2}}\, dx &= \int \frac{1}{\sqrt{1 - (x + 2)^2}}\, dx \\ &= \arcsin(x + 2) + c.\end{aligned}$$

44 b)

$$\begin{aligned}\int \frac{1}{\sqrt{-80 + 18x - x^2}}\, dx &= \int \frac{1}{\sqrt{1 - (x - 9)^2}}\, dx \\ &= \arcsin(x - 9) + c.\end{aligned}$$

45 a)

$$-35 - 24x - 4x^2 = 1 - 4(x + 3)^2,$$

so $p = 1$, $q = 4$ and $r = 3$.

45 b)

Using $2x+6=\sin u$, $2\,dx=\cos u\,du$:

$$\begin{aligned}\int \frac{1}{\sqrt{-35-24x-4x^2}}\,dx &= \frac{1}{2}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du \\ &= \frac{1}{2}u + c = \frac{1}{2}\arcsin(2x+6)+c.\end{aligned}$$

46 a)

$$\begin{aligned}\int \frac{1}{\sqrt{-63-32x-4x^2}}\,dx &= \int \frac{1}{\sqrt{1-(2x+8)^2}}\,dx \\ &= \frac{1}{2}\arcsin(2x+8)+c.\end{aligned}$$

46 b)

$$\begin{aligned}\int \frac{1}{\sqrt{-99+100x-25x^2}}\,dx &= \int \frac{1}{\sqrt{1-(5x-10)^2}}\,dx \\ &= \frac{1}{5}\arcsin(5x-10)+c.\end{aligned}$$

47 a)

$$-27+24x-4x^2 = 9-4(x-3)^2,$$

so $p=9$, $q=4$ and $r=-3$.

47 b)

Using $\frac{2}{3}x-2=\sin u$, $dx=\frac{3}{2}\cos u\,du$:

$$\begin{aligned}\int \frac{1}{\sqrt{-27+24x-4x^2}}\,dx &= \frac{1}{2}\int \frac{\cos u}{\sqrt{1-\sin^2 u}}\,du \\ &= \frac{1}{2}u + c = \frac{1}{2}\arcsin\left(\frac{2}{3}x-2\right)+c.\end{aligned}$$

48 a)

$$\begin{aligned}\int \frac{1}{\sqrt{-11-36x-9x^2}}\,dx &= \frac{1}{3}\int \frac{1}{\sqrt{1-(\frac{3x+6}{5})^2}}\,dx \\ &= \frac{1}{3}\arcsin\left(\frac{3x+6}{5}\right)+c.\end{aligned}$$

48 b)

$$\begin{aligned}\int \frac{1}{\sqrt{3+10x-25x^2}}\,dx &= \frac{1}{5}\int \frac{1}{\sqrt{1-(\frac{5x-1}{2})^2}}\,dx \\ &= \frac{1}{5}\arcsin\left(\frac{5x-1}{2}\right)+c.\end{aligned}$$

49)

$$\begin{aligned}\int_2^{9/2} \frac{1}{\sqrt{21+4x-x^2}}\,dx &= \int_2^{9/2} \frac{1}{\sqrt{25-(x-2)^2}}\,dx \\ &= \left[\arcsin\left(\frac{x-2}{5}\right)\right]_2^{9/2} \\ &= \arcsin\frac{1}{2}-\arcsin 0 = \frac{\pi}{6}.\end{aligned}$$

So $k=\frac{1}{6}$.

50)

$$\begin{aligned}V &= \pi\int_0^{20} y^2\,dx = \pi\int_0^{20}\frac{1}{400+x^2}\,dx \\ &= \frac{\pi}{20}\left[\arctan\left(\frac{x}{20}\right)\right]_0^{20} \\ &= \frac{\pi}{20}\left(\frac{\pi}{4}-0\right) = \frac{\pi^2}{80}.\end{aligned}$$

51)

$$\begin{aligned}\int_{2/5}^{\infty}\frac{1}{4+25x^2}\,dx &= \left[\frac{1}{10}\arctan\left(\frac{5}{2}x\right)\right]_{2/5}^{\infty} \\ &= \frac{1}{10}\left(\frac{\pi}{2}-\arctan 1\right) = \frac{1}{10}\left(\frac{\pi}{2}-\frac{\pi}{4}\right) = \frac{\pi}{40}.\end{aligned}$$

So $k=\frac{1}{40}$.